

Neural Scaling Laws for Customer Choice Prediction

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Working paper

Abstract

Neural scaling laws are empirical relationships between the predictive performance of neural networks and the scale of training data, model capacity, or computation. They have been observed across multiple domains, including language modeling, computer vision, and speech recognition. Yet whether comparable scaling laws hold for prediction tasks in operations remains largely unexplored. We study neural scaling laws relating predictive loss to training-data size in customer choice prediction, a central problem in operations that underlies assortment planning, pricing, and product design. In particular, we train a neural network on an Expedia hotel-search interaction dataset to predict customer bookings across a range of training-set sizes. We establish a power-law relationship between prediction loss and training-set size, and provide an explicit functional form for this relationship. We then examine how this law changes when real choice data are supplemented with synthetic choice data generated by a large language model (LLM). We show that synthetic data improve prediction when real data are scarce, but their benefits shrink as real-data scale grows and can reverse when too much synthetic data is added. We provide an explicit joint scaling law that captures how predictive loss changes with the amount of real and synthetic choice data. Together, these laws can help firms anticipate the predictive returns to additional choice data and assess when and how much synthetic data are likely to improve prediction.

1 Introduction

Neural scaling laws are empirical relationships between the predictive performance of neural networks and the scale of training data, model capacity, or computation (Hestness et al., 2017; Kaplan et al., 2020; Hoffmann et al., 2022). They have been established across multiple domains, including language modeling (Kaplan et al., 2020), computer vision and speech recognition (Hestness et al., 2017), neural machine translation (Ghorbani et al., 2021), and recommendation systems (Ardalani et al., 2022). Depending on the setting, these laws may take the form of a power law (Hestness et al., 2017; Kaplan et al., 2020), sigmoidal scaling in which performance improves slowly at first, then rapidly, and finally approaches a limit (Alabdulmohsin et al., 2022; Jones, 2021), or a broken power law whose scaling rate changes as scale increases (Caballero et al., 2023), among others. By quantifying

returns to scale, these laws help anticipate the value of additional data or computational resources. Yet neural scaling laws for prediction problems in operations remain largely unexplored.

We focus in this paper on customer choice prediction, in which the goal is to predict which product or service a customer selects from a set of available alternatives. Customer choice prediction is a central problem in operations that underlies several important decision-making problems, including assortment planning (van Ryzin and Mahajan, 1999), pricing (Li and Huh, 2011), and product design (Kohli and Krishnamurti, 1987). Recent work has shown that neural networks can capture complex choice patterns and improve predictive accuracy (Rosenfeld et al., 2020; Aouad and Désir, 2025). However, how the predictive performance of such neural choice models scales with the amount of training data remains poorly understood.

Generating synthetic data provides one way to expand training sets without collecting additional real observations. Large language models (LLMs) have recently been used to generate synthetic data across a wide range of applications. For example, LLM-generated data have been used to train models for text classification (Ye et al., 2022), fine-tune language models (Wang et al., 2023), and supplement real customer choice data (Wang et al., 2026; Shapira et al., 2026). This raises the related question of how the predictive performance of neural choice models scales when training sets combine real and synthetic data.

The goal of this paper is to study how the predictive performance of neural customer choice models scales with training-data size and how supplementing real data with synthetic data changes the resulting scaling law. We discuss our contributions in greater detail below.

1.1 Contributions.

Neural scaling laws for hotel booking prediction. We study how the predictive performance of a neural network model for hotel booking prediction changes as the amount of observed customer-choice data increases. In particular, we train a feed-forward neural network to predict whether a customer books a hotel, given the customer’s search context and hotel characteristics. We vary the training-set size from 30 thousand to 24 million observed interactions, drawn from a pool of 27.5 million historical Expedia interactions, and evaluate each trained model on a held-out test set of realized customer choices. We find that predictive loss follows a remarkably regular power-law relationship with training-data size, of the form

$$L(H) = L_{\infty} + AH^{-\alpha}, \tag{1}$$

where $L(H)$ is the predictive loss for a training set of size H , and L_{∞} , A , and α are constants. This relationship mirrors power-law data scaling observed in other learning domains, including language modeling (Kaplan et al., 2020), image classification, and speech recognition (Hestness et al., 2017). This scaling law could be particularly useful for firms seeking to anticipate the predictive gains from collecting additional customer-choice data and assess how those gains diminish as their data scale grows.

How synthetic choice data change neural scaling laws. We next study how neural scaling laws change when real choice data are supplemented with synthetic choice data. In particular, we train the same feed-forward neural network on different combinations of H real-labeled and S synthetic-labeled observations, varying both quantities independently, and evaluate each trained model on the same test set of realized customer choices. The synthetic-labeled observations are drawn from a pool of 9.97 million additional Expedia hotel-search interactions whose synthetic booking labels were generated through zero-shot prompting with the Gemma 4 LLM model (`google/gemma-4-26B-A4B-it` (Gemma Team, 2026)). We find that synthetic data improve prediction when real data are scarce, but their benefits shrink as real-data scale grows and can reverse when too much synthetic data is added. We capture this behavior through a joint scaling law of the form

$$L(H, S) = L_\infty + (1 - \rho)^p [L(H) - L_\infty] + \rho^q [L_S - L_\infty], \quad (2)$$

where $\rho = S/(H + S)$, and L_S , p , and q are constants. The resulting law quantifies how the value of synthetic choice data depends on the amount of real choice data already available, and could help firms assess when and how much synthetic data are useful for improving prediction.

2 Hotel Booking Prediction: Data, Model, and Evaluation

We consider a hotel-booking prediction task in which we train a neural network to predict whether a customer books a hotel based on the customer’s search context and hotel characteristics. This section describes the data, neural network model, training procedure, and evaluation procedure used throughout the paper.

2.1 Hotel Booking Data

We use the Expedia Hotel Recommendations dataset (Kan and Kan, 2016). The dataset contains 37,670,293 hotel-search interactions. Each row represents a customer’s interaction with a hotel in a particular search context. The prediction target equals 1 for interactions that record a booking and 0 for interactions that record a click without a booking, with bookings accounting for 7.97% of the observations. The input features describe the timing and channel of the search, the customer’s location, the trip and travel party, and the hotel and destination characteristics. We order all observations chronologically, reserve the final 100,000 for testing and the preceding 100,000 for validation, and use 37,469,874 of the preceding observations to construct the training datasets. Appendix A provides further details on data construction and the chronological split.

2.2 Neural Network Model, Training, and Evaluation

For all our prediction tasks, we train a feed-forward neural network to predict the probability that a hotel interaction records a booking. The network takes 187 input features: 20 categorical features represented by learned embeddings and 167 numeric features. Appendix A.3 provides full details on

feature construction. We combine the embeddings and numeric features and pass them through two hidden layers of 256 units each. The final layer produces a single score, and we apply the sigmoid function to obtain the predicted booking probability. Each hidden layer uses ReLU activation, batch normalization, and dropout with rate 0.1. Appendix B.1 provides full details on the network architecture. We train the network by minimizing binary cross-entropy (BCE), assigning equal weight to every training observation. Appendix B.2 provides the full training protocol. After each training epoch, we compute BCE on the chronological validation set and retain the checkpoint with the lowest validation loss. We then evaluate the selected checkpoint on the chronological test set, using its BCE as our measure of predictive loss. Appendix F reports the computational environment used for neural-network training.

3 A Neural Scaling Law for Hotel Booking Prediction

This section studies how prediction loss scales with the amount of customer-choice data used to train the neural network. We next describe the experimental setup and the scaling law we establish.

Experimental setup. From the 37,469,874 training observations, we use a fixed pool of 27,496,681 observations for this experiment. We vary the training-set size H across 23 values ranging from 30,000 to 24 million observations. Appendix C lists the exact values. For each H , we fit the network three times. Before each run, we shuffle this pool and select H observations. We denote the mean test BCE across these runs by $L(H)$. Repeating the procedure across different samples makes $L(H)$ less dependent on any particular set of training observations.

Results and discussion. Figure 1(a) shows that mean test BCE is remarkably well described by the power-law relationship in Equation (1). Following Goedhart et al. (2023), we estimate the numerical values of L_∞ , A , and α jointly by unweighted nonlinear least squares using the 23 values of $L(H)$, with equal weight assigned to each training-set size. We measure H in millions of observations. The fit yields $\hat{L}_\infty = 0.20248$, $\hat{A} = 0.007361$, and $\hat{\alpha} = 0.6012$, with $R^2 = 0.9914$. Appendix D provides further details on the estimation and fit.

These results indicate that additional customer-choice data produce systematic but diminishing predictive gains. The estimated scaling law can therefore help firms anticipate the value of collecting more customer-choice data as their existing data scale grows.

4 A Joint Scaling Law for Real and Synthetic Choice Data

This section studies how the neural scaling law established in Section 3 changes when real customer-choice data are supplemented with synthetic customer-choice data generated by an LLM. We next describe the synthetic-data generation process and the joint scaling law we establish.

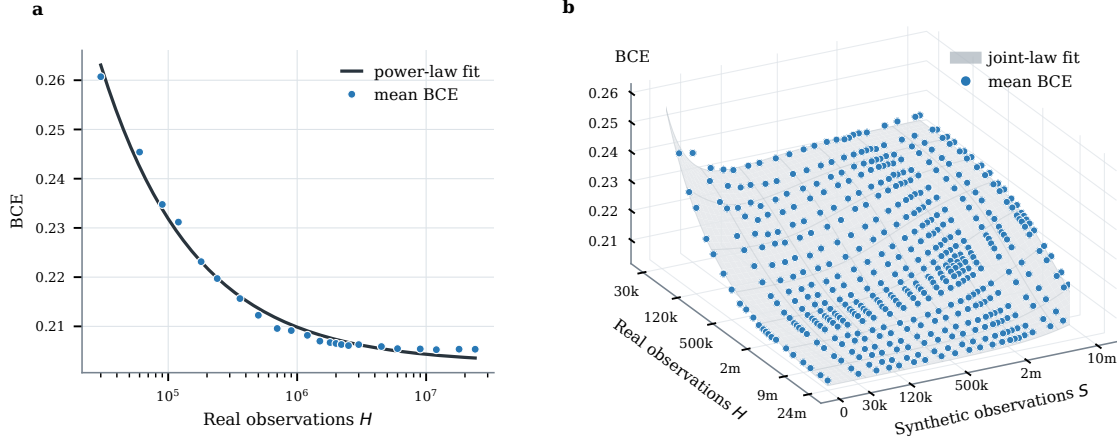


Figure 1: Scaling results.

4.1 LLM-Generated Customer-Choice Data

From the 37,469,874 training observations, we use the remaining fixed pool of 9,973,193 observations that was not used in Section 3 to generate synthetic customer-choice data that we use in this experiment.

In particular, for each observation, we provide Gemma 4 (google/gemma-4-26B-A4B-it; Gemma Team, 2026) with the observation’s search and hotel attributes, together with the overall booking rate and the booking rates for its hotel cluster and destination, and ask it to return a booking probability $\tilde{p}(x) \in [0, 1]$ and a binary booking choice $\tilde{y}(x) \in \{0, 1\}$ for the observation. We use zero-shot prompting, meaning that we do not provide Gemma with labeled examples in the prompt. See Appendix E for the exact prompt we use. Appendix F reports the computational environment used for Gemma label generation.

4.2 Joint Scaling-Law Experiment

Experimental setup. We train the neural network on combinations of H real customer-choice observations drawn from the fixed pool used in Section 3 and S synthetically labeled customer-choice observations drawn from the pool of synthetic customer-choice data generated in Section 4.1. For real observations, the BCE training loss uses the observed binary booking outcome as the target. For synthetic observations, it uses Gemma’s predicted booking probability as the target. Appendix H reports the corresponding results when Gemma’s generated binary booking choice is used instead. We vary H across 23 values ranging from 30,000 to 24 million observations and S across 22 values ranging from 0 to 9,973,193 observations. Appendix C lists the exact grids. For each combination of H and S , we fit the network three times. Before each run, we shuffle the two pools separately and select H real observations and S synthetically labeled observations. We then combine the selected observations and shuffle the resulting training set before fitting the network. Each training set contains $H + S$ observations. We denote the mean test BCE across the three runs by $L(H, S)$.

Results and discussion. When real data are scarce, adding synthetic data can reduce BCE relative to $S = 0$, but only over a range that generally contracts as H grows. At $H = 90,000$, the best observed $S = 120,000$ reduces BCE by 4.16%, and gains persist through $S = 4.5$ million. In contrast, at $H = 500,000$, only $S = 60,000$ improves BCE, by 0.13%. Figure 1(b) shows this pattern, and Appendix G provides the complete results for each fixed value of H .

While, to the best of our knowledge, no standard scaling law describes $L(H, S)$ when real and synthetic customer-choice data vary jointly, Equation (2) fits the empirical surface remarkably well, as Figure 1(b) shows. Equation (2) has a simple interpretation. At $S = 0$, it reduces to $L(H)$. At $H = 0$, it approaches L_S , the nearly constant synthetic-only BCE we observe. Between these boundaries, it allows loss to first fall and then rise as S increases at fixed H . We set $L_S = 0.23427$ to the average of the 21 synthetic-only BCE values, each averaged across three runs, and estimate p and q by unweighted nonlinear least squares over the 506 mean-loss values with $H > 0$, holding the Section 3 law $L(H)$ fixed. The fit yields $\hat{p} = 1.539$, $\hat{q} = 1.378$, and $R^2 = 0.9897$. Appendix G discusses this interpretation and the limiting cases in greater detail. We therefore conjecture that Equation (2) captures the joint scaling of predictive loss with real and synthetic customer-choice data.

The conjectured law could help firms assess whether synthetic choice data are useful at their current real-data scale and how much to add. More broadly, the results show that the predictive value of synthetic data depends on the amount of real data already available.

5 Conclusion and Future Work

We establish an empirical scaling law for neural customer-choice prediction with real data and a joint law for training with real and LLM-generated choice data. Our ongoing work explores several directions for extending our analysis, including (i) testing the scaling laws at larger data scales and across alternative neural architectures; (ii) studying how alternative synthetic-data generation methods, including few-shot prompting and calibration of generated probabilities, affect the joint scaling law; and (iii) extending the analysis to customer-choice prediction tasks beyond hotel booking to examine how the scaling laws change across prediction tasks.

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A Data Construction

A.1 Raw Expedia Data

We use the `train.csv` and `destinations.csv` files from the Expedia Hotel Recommendations dataset (Kan and Kan, 2016). The interaction table `train.csv` contains 37,670,293 rows and 24 columns. Each row records a customer’s interaction with a hotel in a particular search context. Table 1 groups the 24 raw fields by the information they describe. The destination table `destinations.csv` contains a destination identifier and 149 numeric latent features describing similarities among destinations.

Table 1: Raw fields in the Expedia interaction table.

information	raw fields
search	<code>date_time</code> , <code>site_name</code> , <code>posa_continent</code> , <code>channel</code>
customer	<code>user_location_country</code> , <code>user_location_region</code> , <code>user_location_city</code> , <code>user_id</code>
trip and travel party	<code>orig_destination_distance</code> , <code>is_mobile</code> , <code>is_package</code> , <code>srch_ci</code> , <code>srch_co</code> , <code>srch_adults_cnt</code> , <code>srch_children_cnt</code> , <code>srch_rm_cnt</code>
destination	<code>srch_destination_id</code> , <code>srch_destination_type_id</code>
hotel	<code>hotel_continent</code> , <code>hotel_country</code> , <code>hotel_market</code> , <code>hotel_cluster</code>
outcome and count	<code>is_booking</code> , <code>cnt</code>

A.2 Data Transformations

Feature construction. We retain 19 of the 24 raw Expedia columns as input features, excluding `date_time`, `srch_ci`, `srch_co`, `user_id`, and `is_booking` from the inputs. We add 168 new input columns as described in Table 2.

Range restrictions. We restrict `lead_time` to values between 0 and 365 days and `stay_length` to values between 0 and 30 days.

Destination information. We add `d1` through `d149` by joining `destinations.csv` to the interaction data using `srch_destination_id`.

Missing values and standardization. Missing categorical values are assigned the category `__MISSING__`, and categories absent from the training data are mapped to index zero. Missing numerical values are replaced with the corresponding training-data median. Numerical columns are then standardized using their training-data means and standard deviations.

A.3 Input Features and Prediction Outcome

Input Features. Each processed observation contains 187 input features.

The 20 categorical features are `site_name`, `posa_continent`, `user_location_country`, `user_location_region`, `user_location_city`, `is_mobile`, `is_package`, `channel`, `srch_destination_id`,

Table 2: Construction of additional input features.

raw columns	columns added	what the added columns represent
<code>date_time</code>	<code>search_year, search_month, search_dow, search_hour</code>	Year, month, day of the week, and hour when the search occurred
<code>srch_ci</code>	<code>checkin_month</code>	Month in which the hotel stay begins
<code>date_time, srch_ci</code>	<code>lead_time, lead_time_log1p, lead_time_missing</code>	Days between the search and check-in, its logarithmic transformation, and an indicator for missing lead time
<code>srch_ci, srch_co</code>	<code>stay_length, stay_length_log1p, stay_length_missing</code>	Nights between check-in and check-out, its logarithmic transformation, and an indicator for missing stay length
<code>srch_adults_cnt, srch_children_cnt</code>	<code>party_size, has_children</code>	Total number of travelers and whether the party includes at least one child
<code>srch_adults_cnt, srch_rm_cnt</code>	<code>rooms_per_adult, adults_per_room</code>	Rooms requested per adult and adults per requested room
<code>orig_destination_distance</code>	<code>orig_destination_distance_log1p, orig_destination_distance_missing</code>	Logarithmic transformation of the physical distance between the customer and hotel, and an indicator for unavailable distance
<code>cnt</code>	<code>cnt_log1p</code>	Logarithmic transformation of the number of similar hotel-interaction events within the same user session
<code>srch_destination_id, destinations.csv</code>	<code>d1 through d149, destination_features_missing</code>	149 numerical destination descriptors extracted from hotel-review text, and an indicator for missing destination information

`srch_destination_type_id, hotel_continent, hotel_country, hotel_market, hotel_cluster, search_year, search_month, search_dow, search_hour, checkin_month, and has_children.`

The 167 numeric features consist of 5 retained raw columns, 13 constructed columns, and 149 destination features. The 5 retained raw columns are `orig_destination_distance`, `srch_adults_cnt`, `srch_children_cnt`, `srch_rm_cnt`, and `cnt`. The 13 constructed columns are `orig_destination_distance_log1p`, `orig_destination_distance_missing`, `party_size`, `rooms_per_adult`, `adults_per_room`, `lead_time`, `lead_time_log1p`, `lead_time_missing`, `stay_length`, `stay_length_log1p`, `stay_length_missing`, `cnt_log1p`, and `destination_features_missing`. The 149 destination features are `d1` through `d149`.

Prediction Outcome. The prediction outcome is the booking indicator `is_booking`, which equals 1 when the interaction records a booking and 0 when it records a click without a booking.

A.4 Chronological Split

We sort the observations by search timestamp from earliest to latest. The final 100,000 observations form the test set, the immediately preceding 100,000 observations form the validation set, and 37,469,874 of the earlier observations form the training set. The training set supplies the observations used to construct the training datasets. The validation set is used to select model checkpoints, and the test set is used only to report predictive loss. Both holdout sets retain the observed booking outcomes.

B Neural Network Architecture and Training

B.1 Network Architecture

For each categorical feature, the network learns a vector representation for every possible category. For a categorical feature with c distinct values, the embedding dimension is

$$\min\{64, \max\{4, \text{round}(1.6c^{0.56})\}\}.$$

The 20 embedding vectors are concatenated with the 167 standardized numeric features and passed through two hidden layers of 256 units each. Each hidden layer applies a linear transformation followed by ReLU activation, batch normalization, and dropout with rate 0.1. A final linear layer produces a single logit, which is converted into a predicted booking probability using the sigmoid function.

B.2 Training Protocol

Loss. For every training set, we minimize mean binary cross-entropy computed from the network logits, assigning equal weight to every training observation. We use the same loss to evaluate the validation and test sets.

Optimization and checkpoint selection. We train the network using Adam with learning rate 10^{-3} , weight decay 10^{-5} , and a batch size of 4,096. Training proceeds for at most 30 epochs. After every epoch, we compute BCE on the validation set and retain the checkpoint with the lowest validation loss. Training stops after eight consecutive epochs without an improvement in validation BCE. We then restore the selected checkpoint and compute its BCE on the test set.

C Training-Data Grids

Both experiments use the same 23 positive real-data sizes H listed below. The joint experiment additionally includes $H = 0$ and uses the 22 synthetic-data sizes S listed below. We evaluate every pair (H, S) except $(0, 0)$.

grid	values
positive values of H	30,000, 60,000, 90,000, 120,000, 180,000, 240,000, 360,000, 500,000, 700,000, 900,000, 1.2m, 1.5m, 1.8m, 2m, 2.2m, 2.5m, 3m, 4.5m, 6m, 9m, 12m, 18m, 24m
values of S	0, 30,000, 60,000, 90,000, 120,000, 180,000, 240,000, 360,000, 500,000, 700,000, 900,000, 1.2m, 1.5m, 1.8m, 2m, 2.2m, 2.5m, 3m, 4.5m, 6m, 9m, 9,973,193

D Estimation and Fit of the Scaling Law for Observed Customer-Choice Data

Let H_i denote the training-set size indexed by i , for $i = 1, \dots, 23$. Following Goedhart et al. (2023), we fit Equation (1) to the corresponding observed values $L(H_i)$ using unweighted nonlinear least squares, solving

$$(\hat{L}_\infty, \hat{A}, \hat{\alpha}) = \arg \min_{\substack{0 < L_\infty \leq 1, \\ A \geq 0, \alpha \geq 0}} \sum_{i=1}^{23} [L(H_i) - (L_\infty + AH_i^{-\alpha})]^2.$$

Each H_i is measured in millions of observations. Solving this problem yields $\hat{L}_\infty = 0.20248$, $\hat{A} = 0.007361$, $\hat{\alpha} = 0.6012$, and $R^2 = 0.9914$.

Figure 2 compares the observed and fitted values of $L(H)$ across the 23 training-set sizes. The points remain close to the 45-degree line throughout the observed loss range.

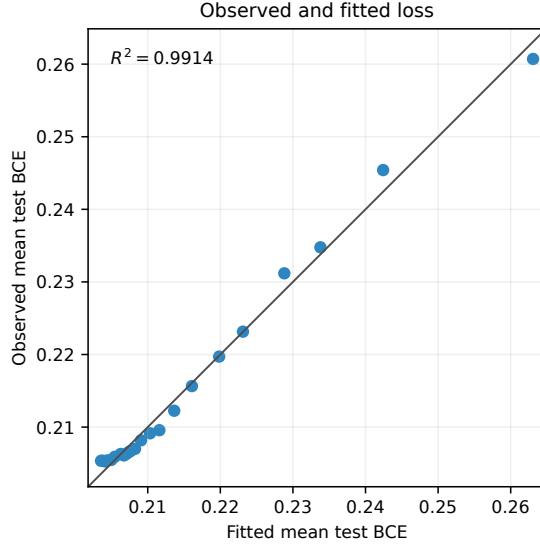


Figure 2: Scaling-law fit. Each point represents one of the 23 training-set sizes, and the 45-degree line indicates exact agreement.

E Gemma Label Generation

We generate the synthetic labels with the Gemma 4 model (google/gemma-4-26B-A4B-it; Gemma Team, 2026) using prompt version `stage2_direct_json_reference_rates_v1`.

Each prompt sent to Gemma contains five real Expedia observations. For each observation, we provide Gemma with 25 search and hotel attributes and three reference booking rates, but not the observed booking outcome. Gemma receives no labeled examples and returns, in input-row order, a booking probability and a separately generated binary booking choice.

The 25 attributes are search month, day of week, and hour; check-in month; lead time and stay length; adult, child, room, and party counts; child, mobile, and package indicators; channel and site;

point-of-sale continent and customer country and region; log destination distance; interaction count; destination type; and hotel continent, country, market, and cluster. The three reference rates are the overall booking rate and the booking rates for the observation’s hotel cluster and destination. We compute these rates from observed booking outcomes among the first 240,000 observations in the fixed ordering used to construct the labeling pool.

The exact prompt sent to Gemma is as follows.

“You are simulating whether an observed Expedia interaction row ends in a booking. Each row is one displayed hotel interaction, not a full choice set. Use only the numeric and categorical fields provided. Do not explain. Return strict JSON only. The first character of your answer must be ‘{’ and the last character must be ‘}’.”

The instruction is followed by the batch-specific text below, where `<batch_rows_json>` is replaced by the JSON serialization of the current batch.

“For each row below, return an object with keys `source_row_id`, `booking_probability`, and `booking_choice`. `booking_probability` must be a number in $[0, 1]$. `booking_choice` must be 0 or 1. Return a JSON object of the form `{"rows": [...]}` where rows is in the same order as the input rows.”

`<batch_rows_json>`

Generation uses temperature zero, a context limit of 32,768 tokens, and a maximum output length of 3,000 tokens.

F Computational Environment

We ran all experiments reported in the paper on a shared high-performance computing cluster. GPU jobs ran on either an NVIDIA H100 NVL GPU with 95,830 MiB of memory or an NVIDIA H100 PCIe GPU with 81,559 MiB of memory, depending on node availability. The two GPU nodes each have 64 physical AMD EPYC 9334 CPU cores.

Each neural-network training job used one H100 GPU and eight CPU cores. Each zero-shot Gemma labeling job used one H100 GPU and 16 CPU cores. Up to four GPU jobs were run concurrently.

The software environment used Python 3.11.15, PyTorch 2.11.0+cu130, CUDA 13.0, cuDNN 9.19.0, vLLM 0.21.0, Transformers 5.8.1, Triton 3.6.0, NumPy 2.2.6, pandas 3.0.3, scikit-learn 1.8.0, and SciPy 1.17.1.

G Detailed Joint Scaling-Law Results

This appendix reports the complete results for the joint scaling-law experiment and provides further details on the interpretation and estimation of Equation (2).

G.1 Complete Observed Surface

The experiment yields 527 mean-loss values. Of these, 506 correspond to 23 positive values of H and 22 values of S , while the remaining 21 correspond to the synthetic-only boundary $H = 0$. Figure 3 reports mean test BCE throughout the positive- H grid. For each H , it also reports the percentage change relative to the corresponding value at $S = 0$,

$$100 \left(\frac{L(H, S)}{L(H, 0)} - 1 \right).$$

Negative values therefore indicate that adding synthetic customer-choice data reduces predictive loss at the given value of H .

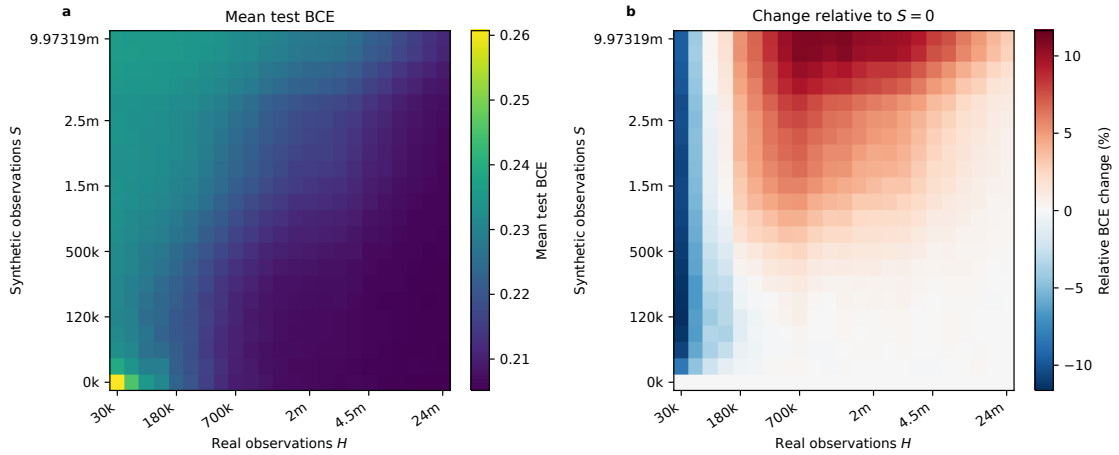


Figure 3: Observed joint loss surface. Panel (a) reports mean test BCE. Panel (b) reports the percentage change in BCE relative to $S = 0$ at each value of H .

Taken together, the two panels reveal two broad patterns. Mean test BCE generally decreases as H increases, whereas the effect of S depends on the amount of real data available. Synthetic data improve prediction primarily at smaller values of H and moderate values of S , but these gains contract as H grows and reverse at sufficiently large values of S .

G.2 Results at Fixed H

Figure 4 shows how predictive loss changes with S at each fixed value of H . Synthetic data provide the largest gains at the smallest real-data scales. As H increases, the range of S over which synthetic data improve prediction generally contracts, and sufficiently large values of S increase BCE.

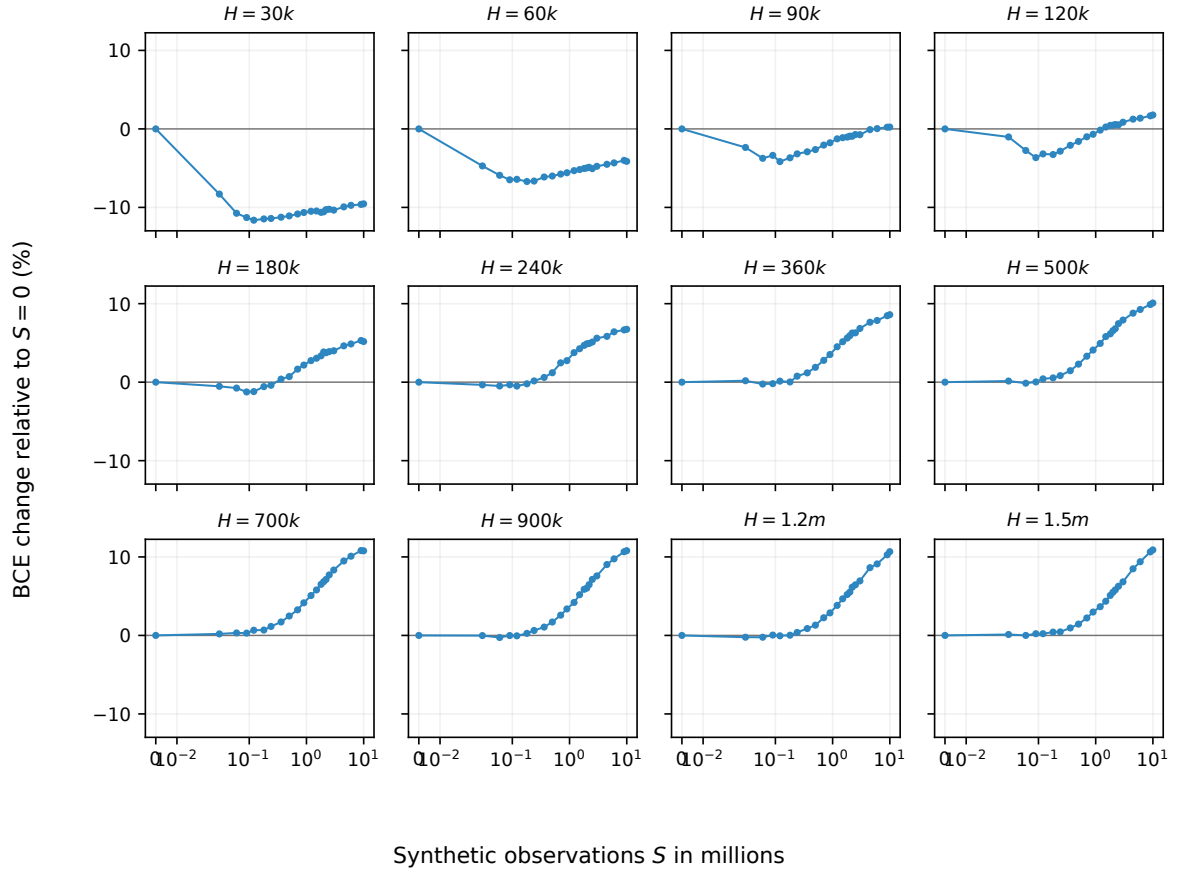
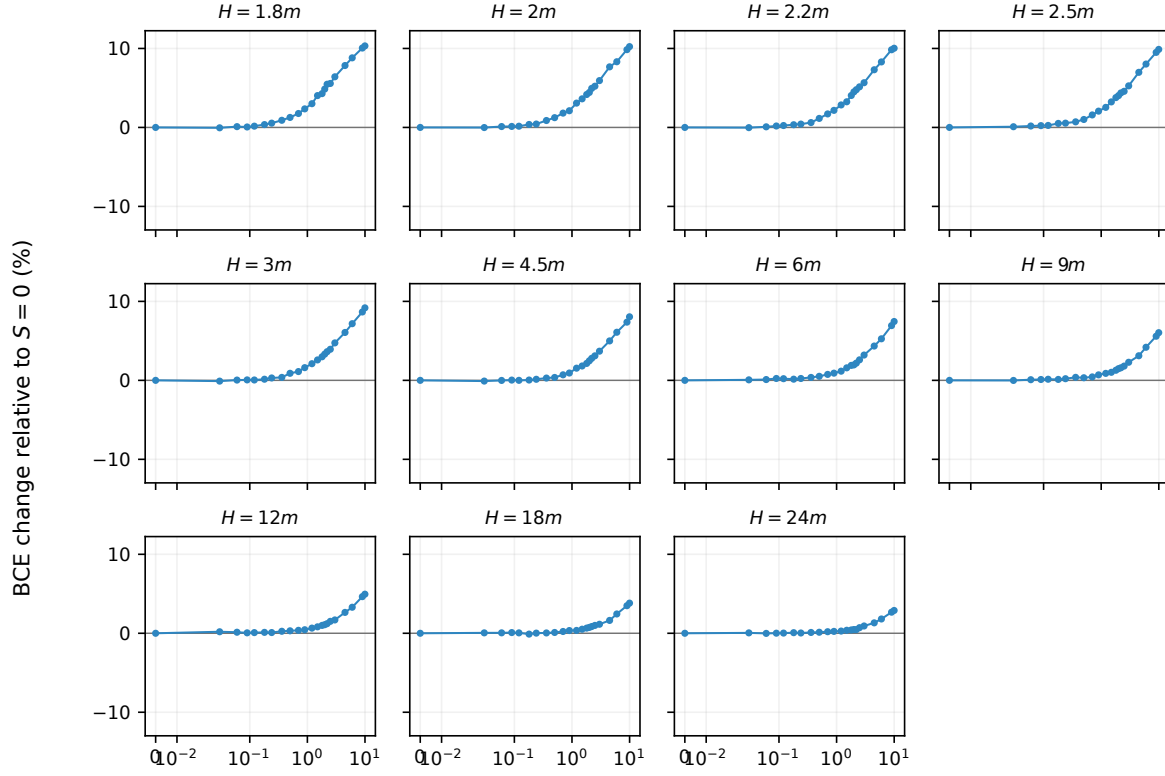


Figure 4: Results at fixed H . Each panel reports the percentage change in BCE relative to $S = 0$ as the amount of synthetic customer-choice data varies.



Synthetic observations S in millions

Figure 4: Results at fixed H (continued).

G.3 Results at Fixed S

Figure 5 shows how predictive loss changes with H at each positive value of S . Predictive loss decreases with H throughout the observed grid. For some fixed values of S , synthetic data reduce BCE when H is small. As H increases, this reduction becomes smaller and can eventually turn into an increase in BCE.

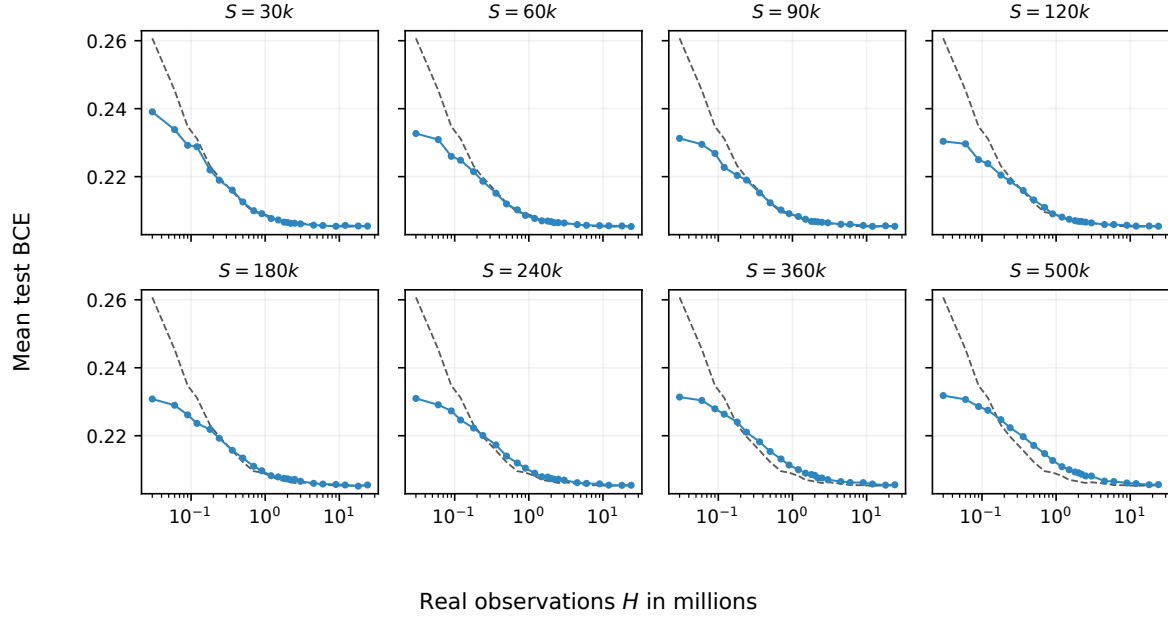


Figure 5: Results at fixed S . Each panel reports mean test BCE as the amount of real customer-choice data varies. The dashed line reports the corresponding loss at $S = 0$.

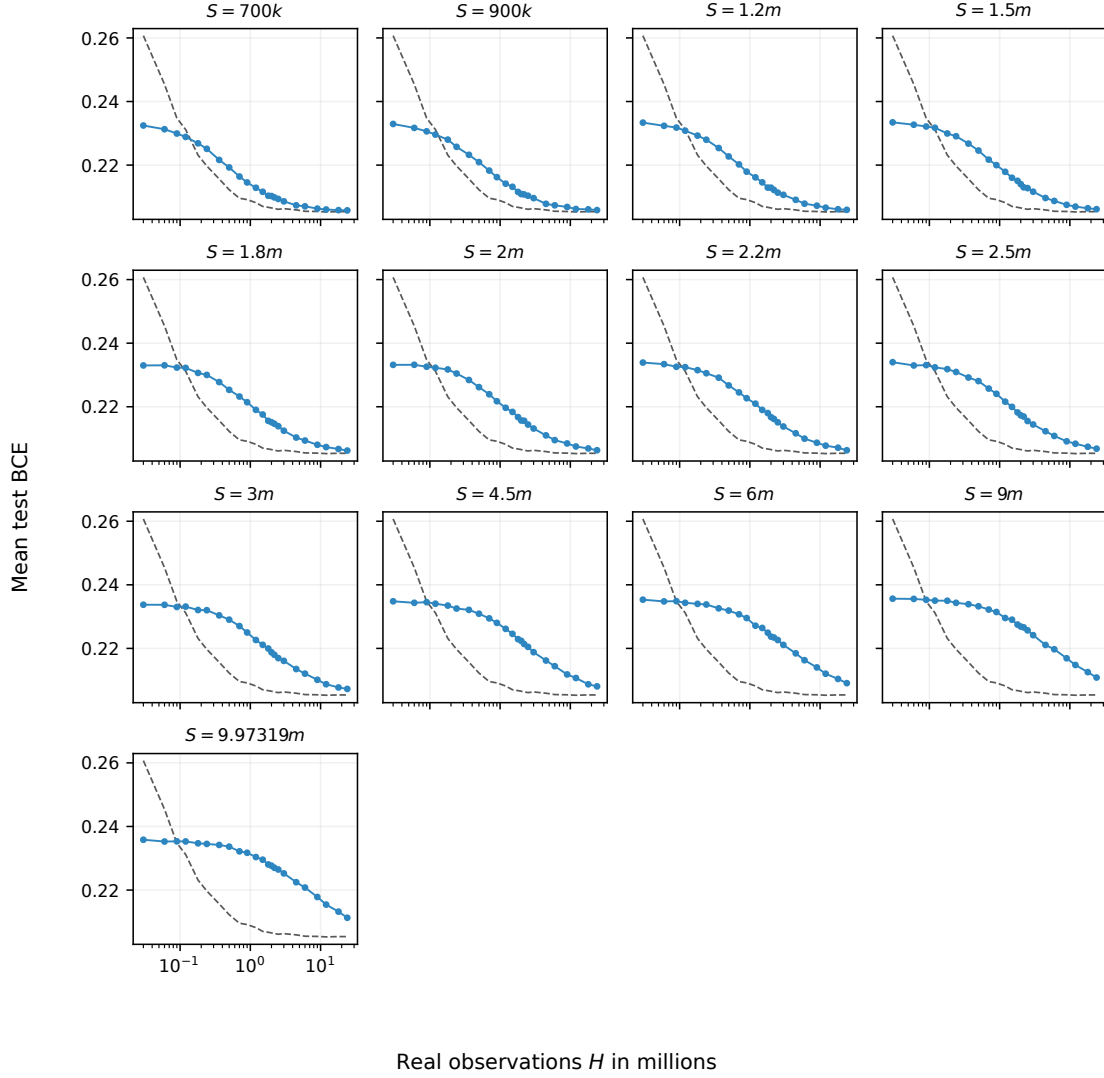


Figure 5: Results at fixed S (continued).

G.4 Boundary Cases

The observed grid contains two boundaries. The $S = 0$ boundary consists of the 23 values used to estimate $L(H)$ in Section 3. The $H = 0$ boundary consists of 21 values of S . Mean test BCE along this second boundary varies within a narrow range around $L_S = 0.23427$, which is the average of these 21 values. Figure 6 reports both boundaries.

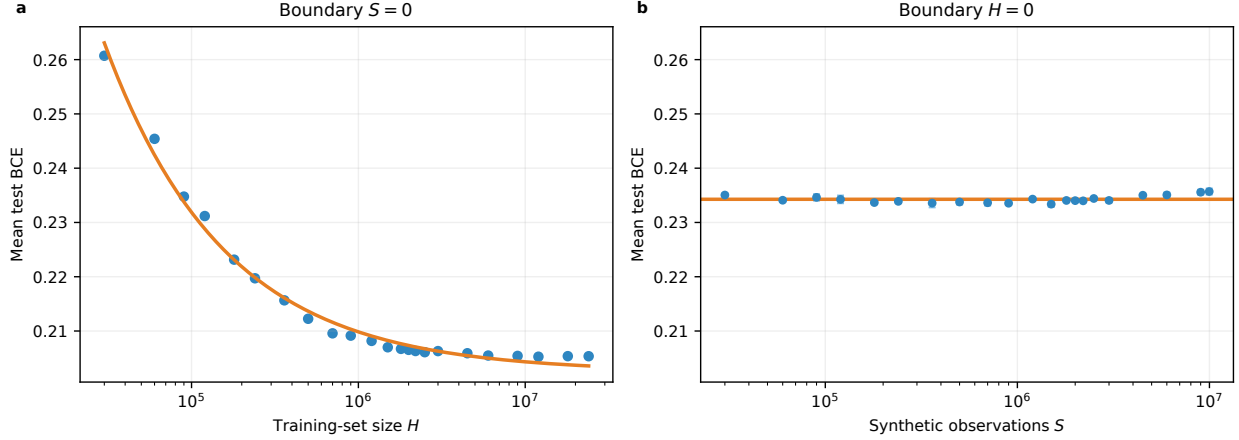


Figure 6: Boundary cases of the observed grid. Panel (a) reports the $S = 0$ boundary and the fitted law $L(H)$. Panel (b) reports the $H = 0$ boundary and the average L_S . Error bars show one standard deviation across the three training runs.

G.5 Estimation and Fit of the Joint Law

Let $\mathcal{G}$ denote the set of 506 observed pairs (H, S) with $H > 0$, and let $\rho = S/(H + S)$ for each pair. We set $L_S = 0.23427$ equal to the average of the 21 observed losses on the $H = 0$ boundary and hold the estimated Section 3 law $L(H)$ fixed. We then fit Equation (2) using unweighted nonlinear least squares, solving

$$(\hat{p}, \hat{q}) = \arg \min_{p, q \geq 0} \sum_{(H, S) \in \mathcal{G}} [L(H, S) - (L_\infty + (1 - \rho)^p [L(H) - L_\infty] + \rho^q [L_S - L_\infty])]^2.$$

Solving this problem yields $\hat{p} = 1.539$, $\hat{q} = 1.378$, and $R^2 = 0.9897$. Figure 7 compares the observed and fitted losses across the 506 pairs.

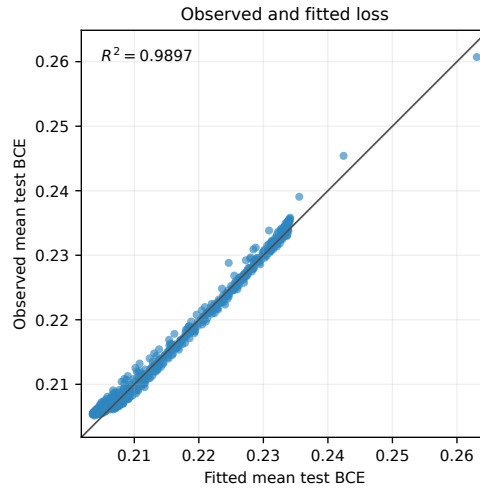


Figure 7: Joint scaling-law fit. Each point represents one of the 506 observed (H, S) pairs, and the 45-degree line indicates exact agreement.

G.6 Interpretation and Limiting Cases of the Joint Law

Equation (2) fits the observed loss surface well, as Figure 7 shows. When $S = 0$, it reduces to $L(H)$, the scaling law for real customer-choice data, as Figure 6(a) shows. At the $H = 0$ boundary, it approaches L_S , the approximately constant loss observed with synthetic data alone, as Figure 6(b) shows. Between these boundaries, the law allows synthetic data to reduce loss when H is small and S is moderate, while sufficiently large values of S can increase loss, as Figures 4 and 5 show.

We interpret this behavior as a trade-off between the amount and quality of training information. When H is small, the network has few observed bookings from which to learn, and a moderate amount of synthetic data provides additional examples of useful choice patterns. The real observations keep the model anchored to realized customer behavior. At $H = 0$, this anchor is absent, so additional synthetic data can help the network reproduce Gemma’s predictions but cannot correct their mismatch with real bookings. As S becomes large relative to H , this mismatch dominates and the initial gains disappear or reverse.

H Results with Generated Binary Booking Choices

The main analysis uses Gemma’s generated booking probabilities as the training targets for synthetic observations. To examine whether the form of the synthetic target matters, we repeat the joint experiment using Gemma’s separately generated binary booking choices as targets. We use the same real and synthetic observation pools, the same 23 values of H , the same 22 values of S , and the same model, training, and evaluation procedures. For each target, we compare the mean test BCE at each (H, S) with the corresponding mean test BCE at $S = 0$ for the same value of H .

Figure 8 replicates the fixed- H analysis in Figure 4 using the generated binary booking choices as synthetic training targets. Each panel reports the percentage change in BCE relative to $S = 0$ as S varies.

Figure 9 replicates the fixed- S analysis in Figure 5 using the generated binary booking choices. Each panel reports mean test BCE as H varies for one positive value of S , together with the corresponding $S = 0$ loss.

Although the response curves in Figures 8 and 9 have broadly similar shapes to those in Figures 4 and 5, the generated binary booking choices provide almost no predictive gain. Only 8 of the 483 combinations with $S > 0$ reduce BCE, and the largest observed reduction is approximately 0.18%. The generated booking probabilities preserve graded information about Gemma’s predicted booking likelihood, whereas a binary choice reduces this information to a 0/1 training target. This information loss helps explain why the binary targets perform substantially worse and shows that the way synthetic customer-choice data are generated matters for their predictive value.

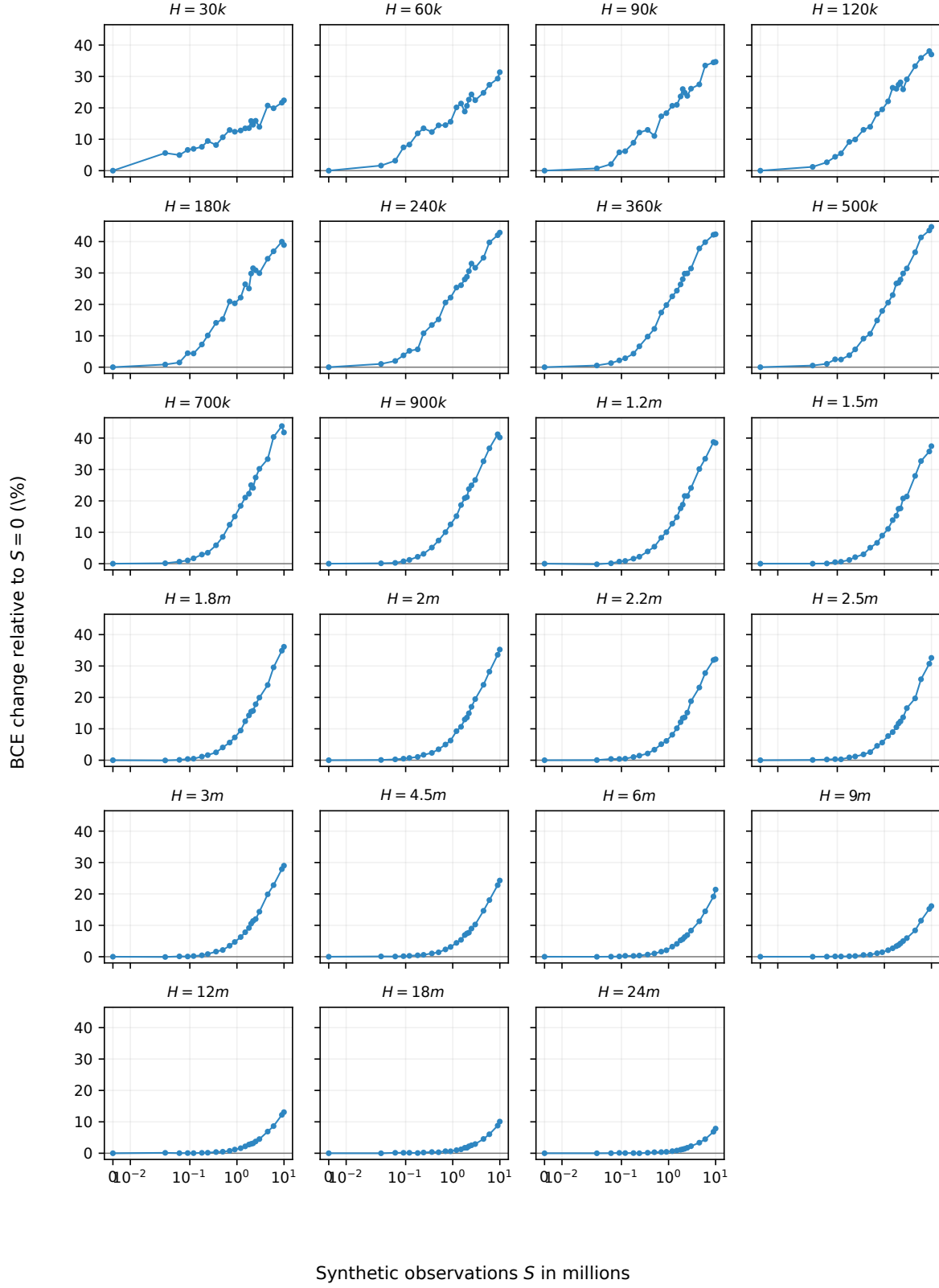


Figure 8: Results with generated binary booking choices at fixed H . Each panel reports the percentage change in BCE relative to $S = 0$ as the amount of synthetic data varies.

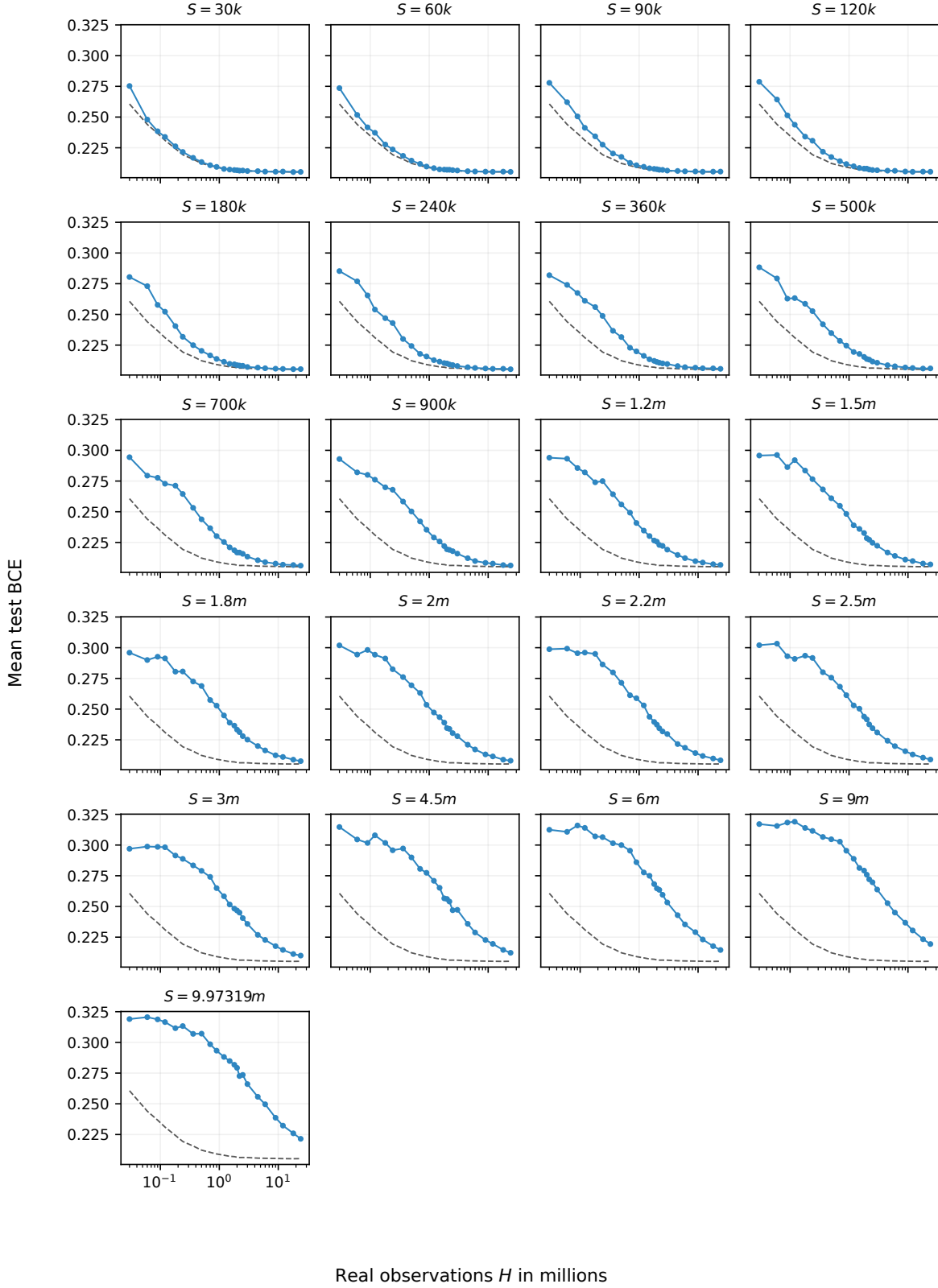


Figure 9: Results with generated binary booking choices at fixed S . Each panel reports mean test BCE as the amount of real data varies. The dashed line reports the corresponding loss at $S = 0$.